



PROFESSIONAL CERTIFICATION

SAP 2000 Training

The SAP 2000 Course introduces structural analysis program for engineers. Learners explore modeling techniques load definitions and analysis methods. Practical training covers static dynamic and nonlinear analysis for bridges buildings and special structures in engineering projects.



30-40

Hours



Certified

Trainers



Industry

Expert



Professional Training | Industry-Recognized Certificate

Dubai, UAE

About Orbit Training Center

Your Gateway to Professional Excellence in Dubai



Who We Are

Established in 2023, Orbit Training Center is a leading professional training provider in Dubai, offering 60+ professional certification courses across diverse fields, including engineering, IT, design, and languages.



Our Benefits

-  Industry recognized certifications
-  Convenient location in the heart of Dubai
-  Affordable fees with multiple payment plans
-  Lifetime post-training support

★Our Features

-  Hands-on learning with real-world projects
-  Expert trainers with industry experience
-  Flexible learning options to fit your schedule
-  State-of-the-art infrastructure & facilities



Our Mission

We deliver outstanding courses, expert teaching, strong career guidance, and placement support to help every student achieve their best results. We stay connected with alumni through networking events, lifelong learning, and ongoing mentorship. We run our operations with smart systems and data-driven decisions to create a smooth, exceptional experience for all.

SAP 2000 Course

It is a way to learn advanced analysis with SAP 2000 tool.

Course Description

This course teaches SAP 2000 for comprehensive structural analysis. Master frame shell and solid modeling pushover and time-history analysis. Projects include bridge design stadium structures and seismic retrofitting for advanced engineering applications.

Duration

30-40

Hours of Training

Format

Classroom / Online

Flexible Option

Key Outcomes

- ✓ Master SAP interface.
- ✓ Create complex models.
- ✓ Apply various loads.
- ✓ Perform nonlinear analysis.
- ✓ Design special structures.
- ✓ Analyze dynamic responses.
- ✓ Generate detailed reports.

Target Audience

- Structure Designers analysis.
- Steel Industry professionals.
- Become Structure Engineer.

Certification

- Certificate of Course Completion
- Industry Recognized Certification
- Trained by Certified Professionals

Prerequisites

- Basic engineering knowledge required.
- Computer skills needed.
- Interest in advanced analysis.

SAP 2000 Course

Begin with SAP basics and model creation.

Intermediate

- Covers all Intermediate Levels
- Precise Geometry Creation
- Basic Parameters & Standards
- Understanding of Tools & Uses
- Intermediate Project Standard
- 1 Project at Completion

Duration
30 Hrs

Professional

- Covers all Professional Modules
- 16 Standard Professional Modules
- Learn with Industry Standard
- Professional Tools and Uses
- Multiple Professional Projects
- 3 Projects at Completion

Duration
35 Hrs

Advanced (Customized)

- Covers All Professional Modules
- 16+ Advance Professional Module
- Co-Ordination and Adv. Parameters
- Shop Drawings & As Built Drawings
- Govt. Standard and Submission
- 4 Projects in Different Categories

Duration
40 Hrs

SAP 2000 Course

Begin with SAP basics and model creation.

01 Exploring SAP2000 Interface & Setup

- > Customize toolbars ribbon display and keyboard shortcuts
- > Configure units preferences decimal precision and colors
- > Master model explorer tables drawing area selection tools
- > Set up grid systems snap controls coordinate references
- > Create reusable steel design templates with standard settings

02 Reference Import Grid System and Story Data Setup

- > Import scale DWG DXF PDF drawings accurately
- > Establish master grid lines and rotated systems
- > Define story heights plinth levels and podiums
- > Assign diaphragm constraints at initial story setup
- > Align geometry and lock reference planes precisely

03 Material Properties Section Definitions and Code Setup

- > Define concrete steel rebar and prestressing materials
- > Specify code-based strength modulus and coefficients
- > Create frame wall slab and tendon sections
- > Import edit international steel shape databases
- > Assign design codes factors and response modifiers

04 Frame Element Modeling Columns Beams and Bracing

- > Draw edit frames using snap and coordinate input
- > Assign sections materials releases and end offsets
- > Apply cardinal points insertion and joint offsets
- > Model inclined haunched and eccentric bracing members
- > Use replicate mirror array for framing efficiency

05 Slab Floor System Modeling with Diaphragm Assignment

- > Generate area objects for various slab systems
- > Assign shell thickness and design strip properties
- > Define rigid semi-rigid flexible diaphragm behaviors
- > Model drops capitals openings and thickened zones
- > Apply controlled auto-mesh with refinement options

06 Shear Wall Core Wall and Lateral System Modeling

- > Draw walls and assign pier spandrel labels
- > Specify shell-thick layered or equivalent properties
- > Model coupling beams with frame or shell behavior
- > Define boundary zones and special seismic detailing
- > Control mesh size for accurate force distribution

SAP 2000 Course

Begin with SAP basics and model creation.

07 Advanced Geometry Modeling for Complex Structures

- > Model transfer girders outriggers and belt trusses
- > Create irregular setbacks stepped and podium floors
- > Implement link gap damper and isolator elements
- > Apply joint body and equation constraints
- > Set staged construction with time-dependent modifiers

08 Comprehensive Load Patterns and Assignment Methods

- > Define dead live partition snow and cladding loads
- > Assign frame area point line and surface loads
- > Include temperature shrinkage creep and settlement cases
- > Generate auto wind seismic loads per code
- > Verify load summation equilibrium and mass source

09 Seismic Parameters Response Spectrum and Modal Analysis

- > Define site class spectral parameters and design category
- > Create code-based or user-defined spectrum functions
- > Set mass participation and orthogonal combination rules
- > Perform eigenvalue analysis and review mode shapes
- > Apply scaling base shear correction and torsion

10 Wind Load Auto-Generation and Manual Pressure Methods

- > Configure directional envelope or user-defined wind cases
- > Specify exposure topography and gust-effect factors
- > Apply pressures on diaphragms and vertical elements
- > Model torsional irregularity with eccentric application
- > Verify base shear overturning and equilibrium checks

11 Load Combinations and Nonlinear Effects and Analysis Control

- > Generate strength serviceability and seismic combinations
- > Include P-Delta large displacement and cracked modifiers
- > Configure nonlinear static time-history pushover cases
- > Set iteration limits convergence and solution controls
- > Execute batch analysis with error monitoring

12 Analysis Results Interpretation and Model Validation

- > Display drifts displacements accelerations and shears
- > Review reactions equilibrium errors and participation ratios
- > Evaluate mode shapes periods and force diagrams
- > Identify soft story torsional and modeling issues
- > Cross-check results with tables and manual verification

SAP 2000 Course

Begin with SAP basics and model creation.

13 Reinforced Concrete Frame Design and Detailing

- › Set preferences overwrites and design combinations
- › Perform beam column brace design per code
- › Review flexural shear axial and interaction ratios
- › Detail longitudinal bars stirrups and confinement
- › Generate rebar tables and export detailing data

14 Shear Wall Coupling Beam and Slab Design

- › Assign piers and run wall flexure shear checks
- › Detail boundary reinforcement and confinement zones
- › Design slabs for punching shear deflection control
- › Generate reinforcement contours and strip-based rebar
- › Check serviceability and long-term deflection limits

15 Steel Frame and Composite Member Design Checks

- › Configure code preferences and load combinations
- › Perform beam column brace truss capacity checks
- › Evaluate buckling lateral-torsional and unity ratios
- › Design composite beams with deck concrete topping
- › Generate connection forces and member summaries

16 Design Optimization Iteration and Reporting Tools

- › Optimize sections using iterative analysis runs
- › Apply stiffness modifiers for cracked behavior
- › Generate detailed design tables and graphical reports
- › Export rebar schedules forces and detailing extracts
- › Prepare input summary assumptions and documentation

Final Projects and Assessment:

- **Assessment:** Project-based evaluation through real-world projects to assess practical skills and application without theory exams.
- **Tailored Curriculum:** Customized to industry standards, covering essential topics for professional development in relevant fields.
- **Hands-on Projects:** Real-world simulations and interactive tasks to build applicable skills and create professional portfolios.
- **Certification Prep:** Targeted review sessions, practice exercises, and strategies for achieving industry certifications.
- **Ongoing Support:** Access to recorded materials, forums, instructor guidance, and career assistance for long-term growth.

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Muppuli Raja



One of great institute for REVIT BIM course. I completed my training with Mr Ahmed. He is one great trainer in this institute. He has very good knowledge for BIM Training. I completed my REVIT BIM training with orbit and it was very nice learning journey for me. And one of best thing is training center is very flexible for classes.



Arun



Best SolidWorks training in Dubai! The instructor made 3D modeling and assembly design easy to understand. The course covered all essential tools I need for product design and engineering work. Best training center for solidworks training. Best Training Center for Best Skills



Harshu CH



My experience at Orbit training center has been the best In every step I learn from there very clearly almost complete my course it's very interesting the trainer is the best professional and friendly ,and the rest of the team keep it up, thanks for supporting me Orbit and all staff...



Ranjan Axalan



I just finished my course in 3ds max. I learned a lot on how to create 3d model. Thank you for the experience and for the teaching method it help me to understand easily and grow my knowledge. I will use my learnings to my future company. Highly Recomanded!



Numan



I had good support from trainer Mr Ahmed and the other staff. They would schedule my BIM classes as per my convenience and would retake classes if I required. They always answer my calls when I need assistance. Not only that the trainer would go a mile a ahead by helping me with my work related issues like a true mentor.



Lelanie Fernandez



Overall, I highly recommend the Fusion 360 training at Orbit Center, especially with Ahmad as the instructor. This course has exceeded my expectations, and I am learning a lot. I would rate it 5 out of 5 stars!

Student Google Reviews

Hear What Our Graduates Say About Their Experience



Aman Bakhtiyar



I had a great learning experience at Orbit Training Center. The trainers are knowledgeable, supportive, and always ready to help. The clear and practical teaching methods really helped improve my skills and confidence. I highly recommend Orbit Training Center to anyone looking to learn AutoCAD, Revit, and other professional skills.



Sana Zahid -



I recently completed the online Digital Marketing course with Orbit Training Center, and it was a great experience! The lessons were clear, practical, and easy to follow. The trainers were knowledgeable and supportive throughout the course. I've learned a lot that I can apply to my work here in Belgium. Highly recommend Orbit Training Center to anyone looking to improve their digital marketing skills!



Cleavon Henderson



Orbit Training Center offers outstanding CAD classes that are easy to understand, even as a first-year mechanical engineering student. The instructors simplify complex concepts and use practical examples, making the learning process engaging and effective. Whether you're new to CAD or improving your skills, Orbit Training Center provides a supportive and enriching environment. Highly recommended!



Danish Khan



Just finished Primavera P6 training at Orbit Training Center and it was fantastic! The trainer made complex scheduling concepts easy to understand with real-world examples. Got hands-on experience with project planning, resource allocation, and progress tracking. The course was well-structured and I feel confident using Primavera P6 for my projects now. Great value and would definitely recommend to other project managers!



500+ Students Trained

Join our growing community of professionals

95%

Satisfaction Rate

60+

Courses Offered

4.6

Average Rating



GET STARTED TODAY

Get in Touch with Us

At Orbit Training Center, we offer a diverse range of professional courses designed to equip you with the latest industry skills and knowledge. Our comprehensive programs cover in-demand fields such as Interior Design, Computer Science, Graphic Design, Digital Marketing, Solar Energy, and specialized software training including STAAD Pro, Revit Architecture, Tekla Structures, and more. Each course is crafted by industry experts to provide hands-on learning, real-world applications, and certification preparation.

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Free

Demo Session



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